MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **WEEKLY TEST – 06** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 8: Introduction to Trigonometry** | **MM: 20**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units where applicable.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective Type Questions

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Q1. If $\sqrt{3}\tan 2\theta - 3 = 0$, where 2θ is an acute angle, then the value of θ is: [1]

(A) 15° (B) 30° (C) 45° (D) 60°

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 30°

Justification: $\tan 2\theta = \dfrac{3}{\sqrt{3}} = \sqrt{3} = \tan 60^\circ \implies 2\theta = 60^\circ \implies \theta = 30^\circ$.

Q2. The value of $\cos 0^\circ \cdot \cos 1^\circ \cdot \cos 2^\circ \cdots \cos 90^\circ$ is: [1]

(A) 0 (B) 1 (C) -1 (D) $\dfrac{1}{2}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 0

Justification: Since $\cos 90^\circ = 0$ is a factor in the product, the entire product evaluates to 0.

Q3. If $\sin(A - B) = \dfrac{1}{2}$ and $\cos(A + B) = \dfrac{1}{2}$ with $0^\circ < A + B \leq 90^\circ$, then B is: [1]

(A) 15° (B) 45° (C) 30° (D) 60°

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 15°

Justification: $A - B = 30^\circ$ and $A + B = 60^\circ$. Subtracting gives $2B = 30^\circ \implies B = 15^\circ$.

Q4. The value of $\dfrac{\sin^2 30^\circ + 4\cot^2 45^\circ - \sec^2 60^\circ}{\csc^2 30^\circ}$ is: [1]

(A) $\dfrac{1}{16}$ (B) $\dfrac{1}{4}$ (C) $\dfrac{3}{16}$ (D) 1

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $\dfrac{1}{16}$

Justification: Numerator $= \left(\dfrac{1}{2}\right)^2 + 4(1)^2 - (2)^2 = \dfrac{1}{4} + 4 - 4 = \dfrac{1}{4}$. Denominator $= 2^2 = 4$. Quotient $= \dfrac{1}{16}$.

Q5. **Assertion (A):** If $\cos A + \cos^2 A = 1$, then $\sin^2 A + \sin^4 A = 1$.
Reason (R): $\cos A = 1 - \cos^2 A = \sin^2 A$, hence $\sin^4 A = \cos^2 A$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: Substituting $\sin^2 A = \cos A$ and $\sin^4 A = \cos^2 A$ from (R) gives $\sin^2 A + \sin^4 A = \cos A + \cos^2 A = 1$. Both are true and (R) explains (A).

SECTION B: Short Answer Type-I

MATHS MANTHAN | (1 × 2 = 2 Marks)

Q6. If $\sec\theta + \tan\theta = 7$, find the value of $\sec\theta - \tan\theta$ and hence calculate $\cos\theta$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Use $\sec^2\theta - \tan^2\theta = 1$ [1 Mark]

$$(\sec\theta + \tan\theta)(\sec\theta - \tan\theta) = 1 \implies 7(\sec\theta - \tan\theta) = 1 \implies \sec\theta - \tan\theta = \frac{1}{7}$$

Step 2: Find $\cos\theta$ [1 Mark]

Adding $\sec\theta + \tan\theta = 7$ and $\sec\theta - \tan\theta = \frac{1}{7}$ gives $2\sec\theta = \frac{50}{7} \implies \sec\theta = \frac{25}{7}$.

$$\boxed{\sec\theta - \tan\theta = \mathbf{\frac{1}{7}}, \qquad \cos\theta = \frac{1}{\sec\theta} = \mathbf{\frac{7}{25}}}$$

Q7. If $x = a\sec\theta + b\tan\theta$ and $y = a\tan\theta + b\sec\theta$, prove that $x^2 - y^2 = a^2 - b^2$. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Compute x^2 and y^2 [1.5 Marks]

$$x^2 = a^2\sec^2\theta + b^2\tan^2\theta + 2ab\sec\theta\tan\theta, \qquad y^2 = a^2\tan^2\theta + b^2\sec^2\theta + 2ab\sec\theta\tan\theta$$

Step 2: Subtract and Apply $\sec^2\theta - \tan^2\theta = 1$ [1.5 Marks]

$$x^2 - y^2 = a^2(\sec^2\theta - \tan^2\theta) - b^2(\sec^2\theta - \tan^2\theta) = \boxed{\mathbf{a^2 - b^2} \quad \text{[Hence Proved]}}$$

Q8. If $\sin\theta + \sin^2\theta + \sin^3\theta = 1$, prove that $\cos^6\theta - 4\cos^4\theta + 8\cos^2\theta = 4$. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Rearrange & Square the Given Relation [1.5 Marks]

$$\sin\theta(1 + \sin^2\theta) = 1 - \sin^2\theta = \cos^2\theta \implies \sin^2\theta(1 + \sin^2\theta)^2 = \cos^4\theta$$

Step 2: Substitute $\sin^2\theta = 1 - \cos^2\theta$ & Expand [1.5 Marks]

$$(1 - \cos^2\theta)(2 - \cos^2\theta)^2 = \cos^4\theta \implies (1 - \cos^2\theta)(4 - 4\cos^2\theta + \cos^4\theta) = \cos^4\theta$$

$$4 - 8\cos^2\theta + 5\cos^4\theta - \cos^6\theta = \cos^4\theta \implies \boxed{\cos^{\mathbf{6}}\theta - \mathbf{4}\cos^{\mathbf{4}}\theta + \mathbf{8}\cos^{\mathbf{2}}\theta = \mathbf{4} \quad \text{[Hence Proved]}}$$

Q9. If $\sec\theta + \tan\theta = p$, prove that $\dfrac{p^2 - 1}{p^2 + 1} = \sin\theta$. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Find $\sec\theta - \tan\theta$ Using $\sec^2\theta - \tan^2\theta = 1$ [1 Mark]

$$\sec\theta - \tan\theta = \frac{1}{\sec\theta + \tan\theta} = \frac{1}{p}$$

Step 2: Express $\tan\theta$ and $\sec\theta$ in Terms of p [1.5 Marks]

Subtracting and adding the two equations give $2\tan\theta = p - \frac{1}{p} = \frac{p^2-1}{p}$ and $2\sec\theta = p + \frac{1}{p} = \frac{p^2+1}{p}$.

Step 3: Divide to Obtain $\sin\theta$ [0.5 Mark]

$$\frac{p^2 - 1}{p^2 + 1} = \frac{2\tan\theta}{2\sec\theta} = \frac{\sin\theta/\cos\theta}{1/\cos\theta} = \boxed{\sin\theta \quad \text{[Hence Proved]}}$$

Q10. Cable-Stayed Suspension Bridge Structural Guy Wires

A cable-stayed pedestrian bridge features a central vertical pylon AB of height 30 m standing perpendicular to the horizontal roadway BC. A steel stay cable AC connects the top of the pylon A to an anchor block C on the bridge deck. The cable makes an angle of $\theta = 30^\circ$ with the horizontal roadway. **Based on the above geometric model, answer the following questions:**

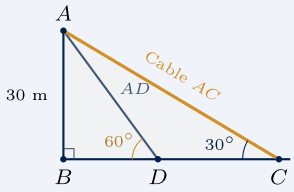
(i) Determine the length of the steel stay cable AC. [1]

(ii) Determine the distance BC from the base of the pylon to the anchor block. [1]

(iii) If a secondary cable AD is anchored at point D between B and C such that $\angle ADB = 60^\circ$, find the length of cable AD and the distance CD between the two anchor points. [2]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Sub-part (i): Length of Stay Cable AC [1 Mark]



$$\sin 30^\circ = \frac{AB}{AC} \implies \frac{1}{2} = \frac{30}{AC} \implies \boxed{AC = \mathbf{60}\text{ m}}$$

Sub-part (ii): Distance BC [1 Mark]

$$\tan 30^\circ = \frac{AB}{BC} \implies \frac{1}{\sqrt{3}} = \frac{30}{BC}$$

$$\implies \boxed{BC = \mathbf{30\sqrt{3}}\text{ m} \approx 51.96\text{ m}}$$

Sub-part (iii): Secondary Cable AD & Anchor Gap CD [2 Marks]

$$AD = \frac{30}{\sin 60^\circ} = \frac{60}{\sqrt{3}} = \mathbf{20\sqrt{3}}\text{ m} \approx 34.64\text{ m}$$

$$BD = \frac{30}{\tan 60^\circ} = \frac{30}{\sqrt{3}} = 10\sqrt{3}\text{ m}$$

$$CD = BC - BD = 30\sqrt{3} - 10\sqrt{3} = \mathbf{20\sqrt{3}}\text{ m} \approx 34.64\text{ m}$$

$$\boxed{AD = CD = \mathbf{20\sqrt{3}}\text{ m} \approx \mathbf{34.64}\text{ m}}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★